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Heart Disease Prediction Using Machine Learning

Objective

The objective of this project was to build a machine learning model capable of predicting the severity of heart disease using the UCI Heart Disease dataset. The task was treated as a multiclass classification problem where the target variable num contains five classes (0–4).

Dataset

The UCI Heart Disease dataset was used for this project. It contains demographic, clinical, and diagnostic features such as age, sex, chest pain type, resting blood pressure, cholesterol level, fasting blood sugar, ECG results, maximum heart rate, exercise-induced angina, and ST depression.

Preprocessing

After the initial data cleaning phase, missing values were handled using appropriate imputation techniques. Median imputation was applied to numerical features such as trestbps and oldpeak, mean imputation was used for thalch, and the most frequent value was used for categorical features such as restecg and exang. Columns with excessive missing values (slope, ca, and thal) were removed from the dataset.

Feature Encoding

Categorical variables were transformed into numerical form. One-hot encoding was applied to dataset, cp, and restecg, while label encoding was applied to sex, fbs, and exang.

Model Training

The dataset was divided into training and testing sets using an 80/20 split with stratification to preserve the class distribution. An **XGBoost** classifier was selected as the primary model for multiclass classification.

Hyperparameter Tuning

GridSearchCV with 5-fold cross-validation was used to optimize the **XGBoost** model. The search included parameters such as n_estimators, max_depth, learning_rate, subsample, and colsample_bytree. The best configuration achieved a cross-validation accuracy of approximately **60.46%**.

Results

The optimized XGBoost model achieved a **test accuracy of approximately 60%**. Additional evaluation was performed using a classification report and confusion matrix. The model performed well on the majority classes but struggled with the minority classes, particularly class 4.

Challenges

The main challenge encountered was **class imbalance**. The higher-severity heart disease classes contained significantly fewer samples, which made them difficult for the model to learn and predict accurately.

Conclusion

A complete end-to-end machine learning pipeline was successfully implemented, including preprocessing, feature encoding, model training, hyperparameter tuning, and evaluation. The final model achieved moderate performance on the original five-class heart disease prediction task, and the results highlighted the impact of class imbalance on multiclass classification accuracy.